LABORATORY ASSIGNMENT № 1

Conditional Statements and Decision-Making in Python

University: Karabakh University
Student Name: Umid Aliyev
Instructor: Şamil Hümbətov
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1. Introduction

This laboratory assignment focuses on conditional statements and decision-making in Python. The main purpose is to practice using if, elif, and else statements to solve programming problems.

2. Objectives

- Apply conditional statements to solve problems.
- Develop decision-making algorithms.
- Practice nested conditions and logical operators.
- Improve problem-solving and programming skills.

3. Task 1: Body Mass Index (BMI) Calculator

Objective

Develop a program that calculates Body Mass Index using the user's weight and height.

Formula

BMI = weight / (height × height)

Requirements

1. Enter weight in kilograms.
2. Enter height in meters.
3. Calculate BMI.
4. Display the BMI value and its corresponding category.

BMI Categories

BMI Range	Category
Below 18.5	Underweight
18.5–24.9	Normal Weight
25.0–29.9	Overweight
30.0 and above	Obese

Implementation

[Describe how your program calculates BMI and uses conditional statements to determine the category.]

Test Results

[Insert your program's input and output.]

4. Task 2: Chinese Zodiac Calculator

Objective

Develop a program that determines the Chinese Zodiac animal based on the user's birth year.

Method

The program uses the remainder of the birth year divided by 12.

The expression year % 12 calculates this remainder.

Zodiac Cycle

Remainder	Animal
0	Monkey
1	Rooster
2	Dog
3	Pig
4	Rat
5	Ox
6	Tiger
7	Rabbit
8	Dragon
9	Snake
10	Horse
11	Sheep

The assignment also states that 1900 was the Year of the Rat.

Implementation

[Explain how your program uses the remainder and conditional statements to identify the animal.]

Test Results

[Insert your test year and the output produced by your program.]

5. Task 3: Birthday Guessing Game

Objective

Create a program that guesses the user's birthday (day of the month) using five predefined number sets.

Method

1. Display the five number sets.
2. Ask whether the birthday is included in each set.
3. Record the user's answers using 1 for Yes and 0 for No.
4. Calculate the birthday based on the responses.
5. Display the guessed date.

The program must not ask the user directly for the birthday date. The result must be calculated entirely from the user's responses.

Implementation

[Describe the logic used to process the five answers and calculate the birthday.]

Test Results

[Include a sample test using the actual number sets and output from your program.]

6. Additional Features

If implemented, describe the following bonus features:

- Input validation.
- Error handling for invalid input.
- A menu that allows users to select a program.
- An option to repeat calculations without restarting the program.

[State which features you actually implemented.]

7. Conclusion

In this laboratory assignment, I practiced using conditional statements and decision-making in Python. The tasks involved calculating BMI, identifying a Chinese Zodiac animal, and implementing a birthday guessing game.

[Add what you learned and any difficulties you encountered while completing the tasks.]

8. References

- Laboratory Assignment 1: Conditional Statements and Decision-Making in Python.
- Python 3 documentation, if used.

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```
print("Hello, it's BMI calculator")
while True:
    giris_ceki = input("\nEnter your weight in kg (Type 'q' to exit): ")
    if giris_ceki.lower() == "q":
        print("Program exited. Thank you!")
        break
    try:
        weight = float(giris_ceki)
        height = float(input("Enter your height (for example, 1.75): "))
        if height <= 0:
            print("Height cannot be zero or negative. Please try again.")
            continue
        BMI = weight / (height * height)
        print(f"Your BMI is: {BMI:.2f}")
        if BMI < 18.5:
            print("Result: Underweight")
        elif 18.5 <= BMI < 24.9:
            print("Result: Normal Weight")
        elif 25.0 <= BMI < 29.9:
            print("Result: Overweight")
        else:
            print("Result: Obese")
    except ValueError:
        print("Please enter valid numbers only!")
```

```
==========================================================
Welcome to the Zodiac Sign Program!

Enter your birth year (Type 'q' to exit): 2009
Your zodiac sign is: Ox

Enter your birth year (Type 'q' to exit):
```

```
Hello, it's BMI calculator

Enter your weight in kg (Type 'q' to exit): 75
Enter your height (for example, 1.75): 1.78
Your BMI is: 23.67
Result: Normal Weight

Enter your weight in kg (Type 'q' to exit):
```

```
Welcome to the Multi-Tool Program!

--- MAIN MENU ---
1. BMI Calculator
2. Chinese Zodiac Finder
3. Birthday Guessing Game
4. Quit
Please choose an option (1-4): 2

--- Chinese Zodiac Finder ---
Enter your birth year (or type 'back' to menu): 2009
Your zodiac sign is: Ox
Enter your birth year (or type 'back' to menu):
```

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```
print("Welcome to the Zodiac Sign Program!")
while True:
  giris = input("\nEnter your birth year (Type 'q' to exit): ")
  if giris.lower() == "q":
    print("Program exited. Thank you!")
    break
  try:
    yas = int(giris)
    zodiak = yas % 12
    if zodiak == 0:
      print("Your zodiac sign is: Monkey")
    elif zodiak == 1:
      print("Your zodiac sign is: Rooster")
    elif zodiak == 2:
      print("Your zodiac sign is: Dog")
    elif zodiak == 3:
      print("Your zodiac sign is: Pig")
    elif zodiak == 4:
      print("Your zodiac sign is: Rat")
    elif zodiak == 5:
      print("Your zodiac sign is: Ox")
    elif zodiak == 6:
      print("Your zodiac sign is: Tiger")
    elif zodiak == 7:
      print("Your zodiac sign is: Rabbit")
    elif zodiak == 8:
      print("Your zodiac sign is: Dragon")
    elif zodiak == 9:
      print("Your zodiac sign is: Snake")
    elif zodiak == 10:
      print("Your zodiac sign is: Horse")
    elif zodiak == 11:
      print("Your zodiac sign is: Sheep")
  except ValueError:
    print("Please enter a valid number!")
```

```
print("Welcome to the Multi-Tool Program!")

while True:
  print("\n--- MAIN MENU ---")
  print("1. BMI Calculator")
  print("2. Chinese Zodiac Finder")
  print("3. Birthday Guessing Game")
  print("4. Quit")

  choice = input("Please choose an option (1-4): ")

  if choice == "1":
    print("\n--- BMI Calculator ---")
    while True:
      giris_ceki = input("Enter your weight in kg (or type 'back' to menu): ")
      if giris_ceki.lower() == "back":
        break
      try:
        weight = float(giris_ceki)
        height = float(input("Enter your height (e.g., 1.75): "))
        if height <= 0:
          print("Height cannot be zero or negative. Try again.")
          continue
        BMI = weight / (height * height)
        print(f"Your BMI result: {BMI:.2f}")
        if BMI < 18.5:
          print("Result: Underweight")
        elif 18.5 <= BMI < 24.9:
          print("Result: Normal Weight")
        elif 25.0 <= BMI < 29.9:
          print("Result: Overweight")
        else:
          print("Result: Obese")
      except ValueError:
        print("Please enter valid numbers only!")

  elif choice == "2":
    print("\n--- Chinese Zodiac Finder ---")
    while True:
      giris = input("Enter your birth year (or type 'back' to menu): ")
      if giris.lower() == "back":
        break
      try:
        yas = int(giris)
        zodiak = yas % 12
        animals = [
            "Monkey",
```

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```
print("Welcome to the Birthday Guessing Game!")
while True:
  start = input("\nDo you want to play? (Type 'quit' to exit): ")
  if start.lower() == "quit":
    print("Program exited. Thank you!")
    break
  set1 = [1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31]
  set2 = [2, 3, 6, 7, 10, 11, 14, 15, 18, 19, 22, 23, 26, 27, 30, 31]
  set3 = [4, 5, 6, 7, 12, 13, 14, 15, 20, 21, 22, 23, 28, 29, 30, 31]
  set4 = [8, 9, 10, 11, 12, 13, 14, 15, 24, 25, 26, 27, 28, 29, 30, 31]
  set5 = [16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31]
  birthday = 0
  print("\nSet 1:\n", set1)
  ans1 = input("Is your birthday included in this set? (yes/no): ")
  if ans1.lower() == "yes":
    birthday += 1
  print("\nSet 2:\n", set2)
  ans2 = input("Is your birthday included in this set? (yes/no): ")
  if ans2.lower() == "yes":
    birthday += 2
  print("\nSet 3:\n", set3)
  ans3 = input("Is your birthday included in this set? (yes/no): ")
  if ans3.lower() == "yes":
    birthday += 4
  print("\nSet 4:\n", set4)
  ans4 = input("Is your birthday included in this set? (yes/no): ")
  if ans4.lower() == "yes":
    birthday += 8
  print("\nSet 5:\n", set5)
  ans5 = input("Is your birthday included in this set? (yes/no): ")
  if ans5.lower() == "yes":
    birthday += 16
  print(f"\nYour birthday is: {birthday}")
```

```
Welcome to the Birthday Guessing Game!

Do you want to play? (Type 'quit' to exit): yes

Set 1:
 [1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31]
Is your birthday included in this set? (yes/no): yes

Set 2:
 [2, 3, 6, 7, 10, 11, 14, 15, 18, 19, 22, 23, 26, 27, 30, 31]
Is your birthday included in this set? (yes/no): no

Set 3:
 [4, 5, 6, 7, 12, 13, 14, 15, 20, 21, 22, 23, 28, 29, 30, 31]
Is your birthday included in this set? (yes/no): n0

Set 4:
 [8, 9, 10, 11, 12, 13, 14, 15, 24, 25, 26, 27, 28, 29, 30, 31]
Is your birthday included in this set? (yes/no): yes

Set 5:
 [16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31]
Is your birthday included in this set? (yes/no): no

Your birthday is: 9

Do you want to play? (Type 'quit' to exit):
```

```
        ]
        print(f"Your zodiac sign is: {animals[zodiak]}")
      except ValueError:
        print("Please enter a valid number!")

elif choice == "3":
  print("\n--- Birthday Guessing Game ---")
  while True:
    start = input("Do you want to play? (yes/back): ")
    if start.lower() == "back":
      break
    if start.lower() != "yes":
      print("Please type 'yes' to play or 'back' to return.")
      continue

    set1 = [1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31]
    set2 = [2, 3, 6, 7, 10, 11, 14, 15, 18, 19, 22, 23, 26, 27, 30, 31]
    set3 = [4, 5, 6, 7, 12, 13, 14, 15, 20, 21, 22, 23, 28, 29, 30, 31]
    set4 = [8, 9, 10, 11, 12, 13, 14, 15, 24, 25, 26, 27, 28, 29, 30, 31]
    set5 = [16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31]
    birthday = 0

    sets_list = [set1, set2, set3, set4, set5]
    powers = [1, 2, 4, 8, 16]

    for i in range(5):
      print(f"\nSet {i+1}:\n", sets_list[i])
      while True:
        ans = input(
            "Is your birthday included in this set? (yes/no): "
        ).lower()
        if ans == "yes":
          birthday += powers[i]
          break
        elif ans == "no":
          break
        else:
          print("Invalid input! Please type 'yes' or 'no'.")

    print(f"\nYour birthday is: {birthday}")

elif choice == "4":
  print("Program exited. Thank you!")
  break
else:
  print("Invalid choice! Please enter a number between 1 and 4.")
```